

Weighted numerator moments in the quasi-Riemann argument

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Abstract

We replace a pointwise numerator bound in the cubic-theta probe by a weighted fourth-moment estimate for its actual smoothed numerator sum. Mellin inversion and the Hecke functional equation expose a shorter plain polynomial, leaving capacity for selected prime factors. A coefficient norm argument controls the original Euler correction, including its ramified error terms, without inserting arbitrary row-dependent coefficients into the moment theorem. Using the original moment parameter $\kappa = 3/4$, the resulting comparison gives $\beta_* \leq 10499/12000 = 0.874916666\dots$ for the original Hecke family and hence for every Dirichlet L -function: none has a zero with $\Re s > 10499/12000$, the pole of the trivial character at $s = 1$ being excluded. The finite exponent checks are exact rational certificates. The complete argument is proved in Lean 4 with Mathlib, on top of the Lean development of the original quasi-Riemann paper. The endpoint theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. A Lake build in the proof package's workspace, with a fresh clone of the pinned fork of the original development, compiled the 2670 modules of that development which the proof imports and all 228 modules of the proof. The Lean FRO Comparator matched the tracker's three template statements for zeta, for all Dirichlet L -functions and for the original Hecke family at $10499/12000$ against definitions compiled from unmodified upstream sources; the con-ron and NanoDa kernels and Lean's own kernel accepted the exported proof. All checks were run by the author on one macOS machine, the three-kernel check without Comparator's sandbox; they are not a tracker receipt. No external human peer review has taken place.

1 Statement, scope, and source inputs

Let β_* be the supremum of $1/2$ and the real parts of nontrivial zeros of the primitive finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$, with arbitrary conductor and height. Principal poles are excluded.

Theorem 1 (Improved zero-free half-plane).

$$\beta_* \leq B := \frac{10499}{12000} = 0.874916666\dots$$

Every finite-order Hecke L -function over $\mathbb{Q}(\sqrt{-3})$, and every Dirichlet L -function including the Riemann zeta function, is therefore zero-free on the open half-plane $\Re s > B$. Only the pole at $s = 1$ of the principal (trivial) characters is excluded.

*GPT-6 Astra (Ultra) was used to discover the proof and perform partial formalization; Opus 5.5 Ultracode was used to complete the formalization and submit the result.

Lean statements. Theorem 1 is proved in Lean by the following unconditional theorems of `Final.lean`. Lean file names in this paper are relative to `formalization/WeightedQRH/` in the accompanying proof package (Section 9).

- `WeightedQRH.beta_le_theta`: `HeckeZeroSupremum.beta` $\leq 10499/12000$.
- `WeightedQRH.hecke_nonzero`: for every `HeckeFamily.Character` χ and every $s \in \mathbb{C}$ with $10499/12000 < \Re s$ and $(s \neq 1 \text{ or } \chi.\text{residue} \neq 1)$, `HeckeFamily.LFunction` χ $s \neq 0$.
- `WeightedQRH.dirichlet_nonzero`: for every modulus q with `[NeZero q]`, every complex Dirichlet character χ modulo q and every s with $10499/12000 < \Re s$ and $\neg(\chi = 1 \wedge s = 1)$, Mathlib’s L -function `DirichletCharacter.LFunction` χ s is nonzero.
- `WeightedQRH.zeta_nonzero`: for every s with $10499/12000 < \Re s$ and $s \neq 1$, Mathlib’s `riemannZeta` is analytic at s and `riemannZeta` $s \neq 0$.

Here `HeckeZeroSupremum.beta` is the definition of the original development [1, 2]. It is the supremum of $1/2$ and the real parts of all zeros with $\Re s > 0$, other than the pole, of `HeckeFamily.LFunction` χ . The supremum runs over every `HeckeFamily.Character` χ , given by a nonzero ideal modulus of the integers of $\mathbb{Q}(\sqrt{-3})$ and a residue character trivial on the global units; χ need not be primitive. This supremum equals β_* . Passing to the primitive character removes finitely many Euler factors $1 - \chi(\mathfrak{p})N\mathfrak{p}^{-s}$, whose zeros lie on $\Re s = 0$. The zeros of a primitive L -function with $\Re s > 0$ are its nontrivial zeros. The Hecke statements concern exactly these finite-order Hecke characters of $\mathbb{Q}(\sqrt{-3})$ in the original definition. They are not a statement of the generalized Riemann hypothesis, and nothing is claimed for other fields.

We use the original physical probe, its zero extensions, common masks, and fixed excluded prime set. No modified zero-free theorem from another refinement is an input. In particular, the moment parameter remains $\kappa = 3/4$ throughout; no lower- κ feedback is used. The starting theorem $\beta_* \leq 7/8$ makes the original hypothesis $\beta_* \leq (1 + \kappa)/2$ available.

The indispensable source results are:

- the original zero-free theorem and `lem:plain` (fourth moment with short prime factors);
- `lem:marked`, `lem:inverse-amplification`, and `prop:detector-witness`;
- `lem:reflected-energy` and `prop:probe-gram`;
- `lem:local-euler` and the exact local formulas underlying `prop:probe-errors`;
- the natural-character reflection and `lem:smooth-calculus`;
- the original contour/tail lemmas and continuation criterion `lem:continuation-criterion`;
- the transfer theorem `prop:hecke-dirichlet-transfer`.

Labels refer to the L^AT_EX source of [1] at the pinned commit given in the bibliography. Our new estimate is proved below; it is not assumed as an additional moment or probe hypothesis. The written proof below cites the source theorems listed above and accepts them as proved in [1]. The Lean proof does not rest on accepting the paper. It uses the Lean formalization of the corresponding source results in the original development [2], together with Mathlib, and its endpoints depend only on the standard axioms (Sections 8 and 9).

2 Geometry and the direct bound

Choose

$$\ell = \frac{167}{1000}, \quad b = \frac{43}{500}, \quad l_x = \frac{747}{2000}, \quad l_y = \frac{919}{2000}, \quad h = \frac{1587}{2000}, \quad \kappa = \frac{3}{4}. \quad (1)$$

Thus $M = l_x + l_y = 1 - \ell$, $b = l_y - l_x$, and $h = 1 - l_x + \ell$. The physical lengths are $X = Z^{l_x}$, $Y = Z^{l_y}$ and $\prod_i P_i = Z^\ell$, with a fixed even number of disjoint prime slots. Let $I_{\eta, \text{modified}}(Z)$ be the finite compensated probe in the source equation **eq:compensated-probe-definition**, evaluated at these lengths, with the same marked-minus-rescaled operations and the same original slot windows. Set

$$S_i(Z) = \sum_{p \in \mathcal{P}_i(Z)} W_i(q_p/P_i) q_p^{-5/6}, \quad A_\eta(Z) = (-1)^K Z^{-\ell/6} \prod_{i=1}^K S_i(Z),$$

and define

$$J_\eta(Z) = \frac{I_{\eta, \text{modified}}(Z)}{c_S A_\eta(Z)}.$$

Here $c_S > 0$ is the fixed scalar from the source equation **eq:shared-principal-data**. The source ray-prime asymptotic gives $|A_\eta(Z)| \asymp (\log Z)^{-K}$ for sufficiently large Z . Both low and high estimates below concern this same normalized probe. The normalized principal signal has exponent

$$C(s) = s - \frac{4+b}{6}.$$

For completeness, the direct estimate at these changed lengths follows from the original estimates without a new cancellation input. For a rescaled subset of total length $v \in [0, \ell]$, put $M' = M - 2v$ and $\ell' = \ell - v$. Then $M' + \ell' - 1 = -3v$. The source reflected-energy proof consequently gives

$$E_{\text{ref}} \leq M' + \frac{(1 + 3\ell' - 2M')_+}{4} + o(1) = M' + \frac{(5\ell - 1 + v)_+}{4} + o(1).$$

After the Gram bound, Cauchy–Schwarz, tuple counting and rescaling, the additional exponent is

$$-v + \frac{(5\ell - 1 + v)_+}{8} \leq 0.$$

Indeed it is zero at $v = 0$ and has slopes -1 and $-7/8$, since $\ell < 1/5$. The required Gram domination is $l_y - \ell - 11b/6 > 0$, and every rescaled length is positive. Therefore the normalized physical probe satisfies

$$|J_\eta(Z)| \ll_{\eta, \epsilon} Z^{l_x/2 + b/12 + \epsilon} = Z^{C(B) + \epsilon}, \quad B = \frac{11}{12} - \frac{\ell}{4}. \quad (2)$$

All strict scale inequalities used here are checked by the certificate in Section 5.

3 Available row counts

Write $U = Z^d$ for a physical row dyad. A dynamic zero bin has $a = (1 + \delta)/2$, with $1/50 < \delta \leq 3/4$ after using $\beta_* \leq 7/8$. The floor bin $\delta = 1/50$ is handled separately. For the physical main prime factors, let $q = \delta x$ be their length-weighted mean amplitude, where $0 \leq x \leq 1/2$; assign amplitude zero to an error slot.

Set $\alpha = 5/6$, $c = 2/9$, and

$$\begin{aligned} D_x &= 3 - (1 + 4c)x, & P_x &= (2 - 4cx)(1 - x), \\ J &= (\alpha - \delta)D_x + \delta P_x, & R &= 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2J}. \end{aligned}$$

The original selected-witness argument gives $\#\mathcal{C} \ll U^{R+\epsilon}(1+T)^A$ for every amplitude/witness subdivision $\mathcal{C}$. Here the plain capacity is exactly $(1-2m)/(6\kappa) = 2(1-2m)/9$. At fixed $\kappa = 3/4$ there is no capacity-replacement loss. To see the count directly, the two short exponents are

$$1 - \delta\{x + (1-x)r\}, \quad 1 - \delta\{2cx + (2-4cx)(t-r)\}.$$

Balance their crossing against the long-witness exponent $1 - \delta + (\alpha - \delta)(t-1)$ by choosing $t = 1 + \delta P_x/(2J)$. This gives the displayed R . The selected inverse capacity is at most $7/37$, and the plain one at most $2/27$. Since $\ell/h > 7/37$, the original whole-slot selection and its small capacity decrements apply. The original zero-capacity arguments are retained. In particular $R \geq 1 - \delta$.

The row count is first applied to the original amplitude class containing *all* physical slots. Its full mean q_{full} is at least the main-slot mean q obtained by setting error-slot amplitudes to zero; both means are divided by the full slot length. For fixed δ , $R(\delta, x)$ is nonincreasing on $0 \leq x \leq 1/2$. Hence the original bound with $R(\delta, q_{\text{full}}/\delta)$ implies the bound with $R(\delta, q/\delta)$ used below. This is the Lean theorem `WeightedQRH.masked_amplitude_class_count` in `HighRows/MaskedCount.lean`. It requires no new moment assertion for a masked slot family.

4 The weighted physical numerator estimate

The useful distinction is between a pointwise bound for $L(w, \chi_{\bullet}(u))$ and its smoothed inverse Mellin transform at the actual scale Y . In the latter, reflection produces a plain polynomial of length at most U/Y , rather than the square-root conductor length used in a generic approximate functional equation. The unused length supports selected prime factors in the existing fourth moment. The next proof includes the Euler correction and every error subset; using only its all-main term would be insufficient.

4.1 A weighted moment for the reconstructed physical numerator

The following is a new analytic amendment to the high estimate. Its inputs are the exact local identities, smooth reflection and scale supremum lemmas, and the fourth moment with short prime factors (Lemma `lem:plain`) of the original manuscript. Those inputs are imported, rather than reproved here. The coefficient bounds, literal Mellin reconstruction, original-numerator fourth moment, and their composition over the amplitude bins are checked in Lean (Section 8); the composition is `WeightedQRH.Numerator.source_integrated_amplitude_classes`, on which `WeightedQRH.Numerator.actual_raw_fullW_input` is built. The argument below requires coefficient estimates for the Euler correction, not merely pointwise bounds for an Euler product.

Write $z_0 = 17/50$, $\kappa = 3/4$, let θ denote the target exponent ($\theta = B$ in Section 5), and assume the already proved global bound $\beta_* \leq 7/8$. Thus the zero-free premise of the imported plain moment holds. This choice of κ is fixed throughout and uses no feedback improvement.

The estimate. Let $U = Z^d$, $Y = Z^{l_y}$, and let $\mathcal{B}$ be a fixed dynamic bin of nonprincipal physical rows u with $q_u \asymp U$. Put $\delta = 2a - 1$, where $51/100 \leq a \leq 7/8$. Let ℓ be the total physical prime-slot length. Suppose

$$\frac{d}{2} < l_y < d, \quad 0 < \frac{2l_y - d}{6\kappa} \leq \ell. \quad (3)$$

For each pointwise amplitude and witness subdivision $\mathcal{C}$ of this bin, suppose the original row-count argument gives

$$\#\mathcal{C} \ll U^{R+\epsilon}(1+T_1)^A, \quad 0 \leq q \leq \delta/2,$$

where $q\ell$ is the sum of slot length times amplitude, assigning amplitude zero to every error slot. Up to arbitrarily small exponent losses and a fixed polynomial in the retained heights, its contribution to the high comparison is bounded by the supremum of

$$E_W(d) = a - \theta + h(z_0 - 1/6) - \frac{l_y}{2} - \frac{\ell}{2} + q\ell + d \left(\frac{3R}{4} + \frac{1}{4} - z_0 \right) - \frac{q(2l_y - d)}{12\kappa}. \quad (4)$$

The supremum is taken after continuing the whole fixed dynamic bin. Individual amplitude sets are not continued separately. In particular, choosing between this estimate and the original high estimate for a whole bin gives $\min\{\sup_q E_{\text{old}}, \sup_q E_W\}$, without requiring an interchange of the minimum and supremum.

Coefficient norms and the local correction. Fix the retained s, z with $\Re s = a + 16e$ and $\Re z = z_0$. Set

$$\rho = \delta/4, \quad \sigma_- = 1/2 - \rho.$$

Then $\sigma_- > 0$ and $\Re s + \sigma_- - 1 \geq \delta/4 > 0$. Both $\Re w = 1/2$ and $\Re w = \sigma_-$ lie in the first Euler region. For a polynomial in Q^{-w} define

$$\left\| \sum_j b_j Q^{-jw} \right\|_\sigma = \sum_j |b_j| Q^{-j\sigma}.$$

The local denominators $1 - D$, $1 - R$, and $1 - V$ in the original manuscript are independent of w and uniformly separated from zero. The factor H_p is a polynomial of degree at most two in Q^{-w} at $p \nmid u$, and at most one at $p \mid u$. The exact defect identity

$$H_p - 1 = \frac{D(V + W - VW) - VW + (1 - V)(1 - W)\mathcal{E}_p}{1 - D}, \quad \mathcal{E}_p = P_p^* + D,$$

therefore gives, by taking absolute coefficients *after* this cancellation,

$$\|H_p - 1\|_{\sigma_-} \ll \begin{cases} Q^{-1-\epsilon_H}, & p \nmid u, \\ Q^{-\epsilon_H}, & p \mid u, \end{cases}$$

for a fixed $\epsilon_H > 0$. The same holds at $\sigma = 1/2$. The convergent prime product and the divisor-product estimate imply that every unselected product, excluding any tuple of distinct slot labels, has absolute coefficient norm $\ll U^\epsilon$ on both lines. These assertions are uniform in the retained imaginary parts.

Write $E_p = G_p + \overline{\chi_p}(u)H_p$. For an error-slot subset I , the quotient-free part of the correction is

$$F_I(w) = \sum_{(p_i)_{i \in I}} \prod_{i \in I} [W_i(q_{p_i}/P_i)q_{p_i}^{z-1}E_{p_i}] \prod_{\substack{p \notin S \\ p \notin \{p_i: i \in I\}}} H_p. \quad (5)$$

All remaining slots retain their actual main factors $-P_i^{z-1/2}Q_i(u; z)$. Equation (5) follows by expanding the holomorphic selected-tuple identity. It does not divide by H_p and does not alter a main-slot mask or coefficient.

At $p \nmid u$, the exact local error cancellation (original equation `old-eq:5.13b`) yields

$$\|E_p\|_\sigma \ll Q^{-s_r} + Q^{-6z_0} + Q^{4-5s_r-6z_0} + Q^{1-\sigma-6z_0} \quad (\sigma \in \{1/2, \sigma_-\}).$$

Each exponent is at most $-1/2$ with a fixed margin. The ideal count therefore bounds the sum over a slot, including Q^{z-1} , by $O(P_i^{z_0-1/2})$.

At $p \mid u$, with $j = v_p(u)$, the main factor is zero and

$$E_p = G_p = \overline{\eta(p)} Q^s (1 - V) \mathcal{E}_p - Q^{-w} H_p.$$

After dividing the slot factor by $Q^{z_0-1/2}$, the non-tail boundary terms of the six-valuation table have the following exponents on $\Re w = \sigma_-$, at $e = 0$:

j	1	2	3	4	5
exponents	$-1 + \rho$	$-\delta$	$-\delta + \rho$ and $-3\delta/2$	$-3\delta/2 + \rho$	$-5\delta/2$

All are negative; positive e improves them. The R, V tails and $-Q^{-w} H_p$ are smaller. The only exception is the strict term

$$-\eta(p)(Q-1)Q^{-s-w}, \quad j \geq 2,$$

whose normalized slot factor costs at most Q^ρ on the left line and at most one on the central line. Ramified labels are divisor-many. For any fixed triangle-expanded tuple, let J consist precisely of its strict ramified labels. They are distinct. Choose the original small-presentation character, whose ideal coefficients agree with the physical row. Let C_u be the product of its primitive conductor norm and full redundant Euler-ideal norm. This combined scale satisfies the original conductor allocation

$$C_u \leq C_S \text{rad}(u) \leq C_* U \prod_{p \in J} Q^{-(j-1)}.$$

Consequently $(C_u/(C_* U))^\rho \prod_{p \in J} Q^\rho \leq 1$. The single conductor deficit is applied to the whole tuple.

Expand $F_I(w) = \sum_k A_{I,u}(k) q_k^{-w}$ and put $B_I = \prod_{i \in I} P_i^{z_0-1/2}$. The preceding estimates prove

$$\sum_k |A_{I,u}(k)| q_k^{-1/2} \ll U^\epsilon B_I, \quad (6)$$

$$\left(\frac{C_u}{C_* U} \right)^\rho \sum_k |A_{I,u}(k)| q_k^{-1/2+\rho} \ll U^\epsilon B_I. \quad (7)$$

Combining them gives the useful weighted form

$$\sum_k |A_{I,u}(k)| q_k^{-1/2} \max \left(1, \frac{C_u q_k}{C_* U} \right)^\rho \ll U^\epsilon B_I. \quad (8)$$

This also explains why a strict ramified error does not consume prime capacity: its forced dilation is paid for by a distinct factor in the conductor deficit.

Reconstructing and reflecting the numerator. For each fixed nonprincipal physical row, move the complete w integral to $\Re w = 1/2$. The numerator is entire; the correction is holomorphic throughout this move, and the Mellin profile has arbitrary decay. The bin itself is independent of w . The absolute coefficient norm just proved and polynomial strip growth justify termwise Mellin inversion:

$$\begin{aligned} & \frac{1}{2\pi i} \int_{(1/2)} Y^{w-1} \widehat{W}_1(w) L^S(w, \chi_u) F_I(w) dw \\ &= Y^{-1/2} \sum_k A_{I,u}(k) q_k^{-1/2} S_{\chi_u}(Y/q_k; W_1). \end{aligned} \quad (9)$$

Here S denotes the original centrally normalized plain sum. Since W_1 has compact support, say in $[r_-, r_+]$, terms with $q_k > r_+ Y$ vanish identically before reflection. Thus all relevant scales lie in a fixed polynomial range.

The imported smooth functional equation replaces $S_{\chi_u}(X; W_1)$ by reflected sums with conjugate character and fixed transformed profiles. Restoring natural zeros follows the original reflection proof. The redundant Euler restoration need not be supported on fixed S : every restoration subset has norm at most the full redundant ideal norm, and every smooth-ideal denominator decreases the reflected scale. Consequently all reflected scales are at most C_u/X , with the combined C_u defined above. The original restoration coefficient mass is subpower in the small modulus. Fixed constants enter C_* . No extra weighted restoration mass or new moving mask is needed.

Use (8) to remove all row-dependent coefficients before applying a moment. It leaves a rowwise supremum of reflected sums of scale at most $C_*(U/Y)T$, multiplied by $T^{-\rho}$, where

$$T = \max(1, C_u q_k / (C_* U)).$$

The imported smooth scale-supremum lemma handles the row-dependent scale using logarithmically many boxes and finitely many derivatives. Annular decomposition of the reflected profile gives the usual arbitrarily small length loss, summable lower annuli and arbitrarily decaying upper tails. Since $q_k \leq r_+ Y$, all main reflected scales are $O(U)$ and the declared lengths stay bounded. Thus no uniform moment assertion for unbounded lengths is being used.

Proposition 1 (Checked original-numerator moment). *Assume $51/100 \leq \beta_*$. (The bound $\beta_* \leq 7/8$ of [1] is available throughout but is not needed here; the Lean statement assumes only the lower bound.) Fix the original weighted source data, its moment loss $t > 0$, a Schwartz profile G , and positive constants λ, ξ, b . For sufficiently large U , let $\mathcal{R}$ be any finite set of nonprincipal physical rows with $U^{1/100} \leq q_u \leq U$. For every row choose $0 < N_u \leq bY$, where $Y > 0$, and its actual combined conductor C_u . Suppose, for a common $m \geq 0$,*

$$m + \xi \leq 2, \quad C_u \leq C_S U^m \frac{Y}{N_u}.$$

Let T be any selected family of original prime slots with nonnegative widths r_i below the certified mesh $\text{fineMesh}(2, 0, 1, 3/4, t/4)$, and let $\Re z_i = 17/50$, $|\Im z_i| \leq H$, $H \geq 0$. Assume

$$2(m + \xi) + \frac{9}{2} \sum_{i \in T} r_i \leq 1 \quad \text{or} \quad T = \emptyset, \quad \left| \prod_{i \in T} Q_i(u; z_i) \right| \leq U.$$

Then there are a fixed integer $J \geq 0$ and $C > 0$ such that

$$\sum_{u \in \mathcal{R}} |S_{\chi_u}(Y/N_u; G)|^4 \left| \prod_{i \in T} Q_i(u; z_i) \right|^2 \leq C(1 + H)^J U^{1+t+\lambda}.$$

The constants and threshold precede all choices of rows, scales, selected slots, and heights. They may depend on the fixed source data, profile and displayed fixed positive constants.

Proof. Use the actual smooth reflection identity with unit-modulus phase and cutoff $U^{\xi/4}$. The certified retained-profile theorem bounds the weighted fourth moment of the retained part by $C_r(1+H)^J U^{1+t+\lambda}$. It uses the positive-capacity source moment when $T \neq \emptyset$ and the unrestricted bounded-length zero-slot source moment otherwise. Both applications preserve the original physical coefficients and selected factors.

For the discarded part, $C_u \leq C_S q_u \leq C_S U$ and $N_u \leq bY$ give a uniform reflected-scale bound $C_u N_u / Y \leq b C_S U$. The original arbitrary-order discarded-tail estimate therefore gives any power saving, uniformly over these rowwise choices. The elementary row count is at most $128U$, and the assumed selected product is at most U . Choosing sufficient tail decay bounds its weighted fourth moment by a constant C_d . Finally, $|r + d|^4 \leq 8(|r|^4 + |d|^4)$, the reflection phase has modulus one, and $(1 + H)^J U^{1+t+\lambda} \geq 1$. The result follows with $C = 8(C_r + C_d)$. The Lean theorem allows the more general fixed bound $C_{\text{slot}} U^{q_{\text{tail}}}$ on the selected product; the displayed statement specializes both constants to one. $\square$

Selecting prime factors inside the fourth moment. Fix an amplitude set $\mathcal{C}$ after the full w integration, at one retained (s, z) and witness tuple. Put

$$L = \ell/d, \quad n_0 = 1 - l_y/d, \quad z_P = \frac{1 - 2n_0}{6\kappa}.$$

For a scale dyad $T = U^v$, allow length $n(v) = n_0 + v$ and define

$$z(v) = \max\left(0, z_P - \frac{2v}{6\kappa}\right).$$

The supply condition gives $z_P \leq L$. Select the largest positive amplitude main slots up to this capacity. If their total length is smaller, select all of them. The original selection lemma, with a sufficiently fine mesh, gives

$$|Q_{\text{selected}}| \geq U^{qz(v)-o(1)}, \quad |Q_{\text{all}}| \leq U^{qL+o(1)}$$

on $\mathcal{C}$. Actual selected length never exceeds $z(v)$; no zero-amplitude factor is required for this lower bound. The reflected numerator has character $\overline{\chi}_{\bullet}(u)$, exactly matching the physical main slots $\overline{\chi}_P(u)$. Conjugating the whole plain/selected-slot product inside its absolute value gives the positive row orientation in the statement of the imported plain moment; it only conjugates the profiles and fixed finite-ray coefficients, and preserves every zero extension. For sufficiently large U in the moderate dyads, a physical row cannot induce a character in the fixed exceptional group Θ , since a good prime of that row ramifies. The finitely many unit or fixed-prime-supported exceptions belong to the separate small-row estimate.

Hölder's inequality and the imported plain moment give

$$\begin{aligned} \sum_{u \in \mathcal{C}} |S_{\tilde{\chi}_u}(n(v)) Q_{\text{all}}(u)| &\ll U^{qL-qz(v)/2+o(1)} (\#\mathcal{C})^{3/4} \left(\sum_u |S_{\tilde{\chi}_u}(n(v))|^4 |Q_{\text{selected}}(u)|^2 \right)^{1/4} \\ &\ll U^{qL-qz(v)/2+3R/4+1/4+o(1)}. \end{aligned}$$

If $z(v) > 0$, the capacity equation is $2n(v) + 6\kappa z(v) = 1$. If $z(v) = 0$, use the original lemma's stronger zero-slot assertion: it gives $U^{1+o(1)}$ for bounded nonnegative plain lengths, without a length restriction. This is the certified **ZeroAt** input, not the weaker downstream detector estimate with exponent $\max(1, 2n)$. Small slack is reserved for annuli and scale boxes. The derivative versions used by the scale supremum have the same exponent and a fixed height cost.

The coefficient weight adds $-\rho v$. Before the capacity is exhausted, the exponent relative to $v = 0$ is

$$\left(\frac{q}{6\kappa} - \rho\right) v \leq -\frac{5\delta}{36}v,$$

because $\kappa = 3/4$, $q \leq \delta/2$, and $\rho = \delta/4$. After capacity is exhausted, its slope is $-\rho$. Thus every dilation dyad is controlled by $v = 0$, with room for the small fixed losses. All error subsets satisfy the same estimate; no lower bound for an error factor is used.

Restoring the main-slot normalizers, the w -integrated bin is bounded by

$$Y^{-1/2} Z^{\ell(z_0-1/2)+q\ell} U^{3R/4+1/4-qz_P/2+o(1)}.$$

The remaining q_u^{-z} factor, reciprocal-denominator estimate and X, Z Mellin weights are those of the original high-bin proof. They give exactly (4). The finite amplitude, witness and error subdivisions are used only in this estimate of the already continued whole bin. Rapid external Mellin decay and the fixed height orders treat the joins and discarded coordinates with the original cumulative height allocation.

Dependence on the row scale. In the positive-capacity range the coefficient of d in (4) is

$$\frac{3R}{4} + \frac{1}{4} - z_0 + \frac{q}{12\kappa}.$$

For the original selected count $R \geq 1 - \delta$ and $\delta \leq 5/6$, this is at least $7/200 > 0$. Thus $d = h$ controls the moderate row range whenever (3) holds throughout that range. The small and far row estimates can remain those of the original proof.

Finally, at fixed R, q the difference from the original high exponent is

$$E_W - E_{\text{old}} = \frac{\delta l_y}{2} + d \left(\frac{1-R}{4} - \frac{\delta}{2} \right) - \frac{q(2l_y - d)}{12\kappa}.$$

The final term is the additional saving supplied by the reconstructed physical numerator and the existing prime-weighted fourth moment. It is independent of any parameter-feedback argument.

5 Exact exponent comparison

Let $z_c = 17/50$. The source pointwise numerator argument gives, relative to $C(B)$,

$$E_{\text{old}}(d) = a - B + h(z_c - 1/6) - al_y - \ell/2 + q\ell + d(R + \delta/2 - z_c). \quad (10)$$

For $1/2 \leq d \leq h$, we have $l_y < d < 2l_y$. The new weighted estimate instead gives $E_{\text{wt}} = E_W$, that is (4) with $\theta = B$ and $z_0 = z_c$:

$$E_{\text{wt}}(d) = a - B + h(z_c - 1/6) - l_y/2 - \ell/2 + q\ell + d \left(\frac{3R+1}{4} - z_c \right) - \frac{q(2l_y - d)}{12\kappa}. \quad (11)$$

The last term is the additional saving from the prime factors in the numerator moment. At a fixed zero bin we choose the source contour when $\delta \leq 1/3$, and the new numerator treatment when $\delta \geq 1/3$. This choice is made for the original dynamic-bin sum before any pointwise amplitude subdivision. It does not choose a contour according to x .

Lemma 1. *For the geometry (1),*

$$\begin{aligned} E_{\text{old}}(h) &\leq -10^{-4} & (0 \leq \delta \leq 1/3, 0 \leq x \leq 1/2), \\ E_{\text{wt}}(h) &\leq -10^{-4} & (1/3 \leq \delta \leq 3/4, 0 \leq x \leq 1/2). \end{aligned}$$

All prime-supply, rescaling, remaining row-range, and Euler-region inequalities needed in this proof hold with the stated strict margins.

Proof. Here is an exact, reproducible finite certificate. Put $A_P = (\alpha - \delta)\delta P_x$, so that the first identity below holds, and form the polynomial T :

$$RJ = (1 - \delta)J + \frac{A_P}{2}, \quad T = \frac{(1 + \delta)(1 - l_y)}{2} - \frac{\ell}{2} + q\ell + \frac{h\delta}{2} - \frac{h}{6} - B.$$

Then the two endpoint numerators are

$$\begin{aligned} JE_{\text{old}}(h) &= JT + h(RJ), \\ JE_{\text{wt}}(h) &= JE_{\text{old}}(h) + \frac{h}{4}[J - RJ] + \frac{\delta(l_y - h)}{2}J - \frac{q(2l_y - h)}{12\kappa}J. \end{aligned}$$

All their coefficients are rational and $J > 0$ on the full rectangle. The script `evidence/certificate/weighted_numerator_certificate.py` in the proof package proves that $-J(E + 10^{-4}) \geq 0$, using the convex-hull property of the Bernstein basis: one rectangle suffices for the first branch, and nine exact rational subrectangles suffice for the second. The endpoints

and lower Bernstein coefficients are stored in `evidence/certificate/weighted_numerator_certificate.json`. No floating-point sign decision is used. Lean proves the two branch inequalities separately, in the extended form used by the proof (Section 8).

The same script checks 32 auxiliary rational comparisons, including

$$\frac{\ell}{h} > \frac{7}{37}, \quad 5\ell - h > 0, \quad 2l_y - h > 0, \quad l_y < \frac{1}{2}, \quad \ell > \frac{2l_y - 1/2}{6\kappa}.$$

The last inequality supplies every numerator capacity on $d \in [1/2, h]$. For $d \leq 1/2$, use the original no-slot count $R = 76/75 - 2\delta/3$; its exponent at $d = 1/2$ is strictly negative at $\delta = 1/50$ and $\delta = 3/4$, hence throughout by affinity. The floor-bin estimate with $R = 1$, $q = 1/100$ is strictly negative at $d = h$. For $d \leq 1/100$ the original absolute small-row comparison has saving

$$-\{h(z_c - 1/6) - l_y/2 + 63/5000\} > 0$$

directly relative to $C(\beta_*)$.

Both endpoint estimates control the full main row interval. The source frequency slope $R + \delta/2 - z_c$ is positive. The new slope is

$$\frac{3R + 1}{4} - z_c + \frac{q}{12\kappa} \geq \frac{7}{16} - \frac{17}{50} > 0,$$

using $\delta \leq 3/4$ and $R \geq 1 - \delta$. All these slopes have a fixed upper bound, so a sufficiently small fixed extension above h preserves the strict endpoint and prime-supply margins. $\square$

6 Euler regions, parameter choices, and continuation

The principal comparison uses the same residue as the source. Its second Euler region extends to $\Re s \geq B$ with $\Re w \geq 19/20$ and $\Re z \geq 33/200$: the good-prime defect exponents $4 - 6\Re s - 6\Re z$ and $1 - \Re s - \Re w - 6\Re z$ remain strictly below -1 . The four principal approximation exponents

$$-\Re s, \quad -6\Re z, \quad 4 - 5\Re s - 6\Re z, \quad 1 - \Re w - 6\Re z$$

are at most $-B$. All local denominators stay uniformly separated from zero. Thus the principal multiplier is $-1 + O(q_p^{-B})$; the normalized correction $H_\eta(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6)$ is holomorphic on $\Re s > B$ and can be made within $1/2$ of one by increasing the fixed excluded-prime cutoff. The principal normalization has a subpower reciprocal. The source contour lemmas with a printed $7/8$ lower endpoint extend to B by precisely these local checks. On the absolute small-row line, choose their first-region buffer below $B - 1/2$.

Suppose for contradiction that $\Delta = \beta_* - B > 0$. First choose a fixed row extension $\zeta > 0$ smaller than the endpoint, supply, and $2l_y - h$ margins. Choose the moment, capacity, reflection truncation, and rounding losses relative to 10^{-4} and Δ . Both the original row counts and the new numerator moment use the original uniform mesh assertion, so one common mesh suffices. Next choose the fixed even number of disjoint slots and their windows. Only after this choice fix the principal approximation loss $0 < \mu < B \min_i \ell_i$. Choose the remaining bin widths, contour buffers, normalization losses and detector losses small relative to the already fixed margins and μ . The uniform coefficient norm and reflected scale arguments in the new lemma consume arbitrarily small losses; they do not change this order of choices.

All these real choices precede the target character. For each target choose its fixed arithmetic data, cutoff, and finite internal seminorm orders. Finally use the source's late height closure: choose the positive height exponent, then the external tail order, then the threshold. The new moment/reflection estimates have the same finite-seminorm and polynomial-height form as the cited source lemmas. To ensure that the new Mellin identity uses the full w line, the checked transport first uses the original physical-row-to-central-cube theorem, with $\Re s = a + 16e$,

$\Re w = 1 - a - 6e$ and $\Re z = z_c$. Extend only the truncated w interval to its entire vertical line; the original polynomial kernel bounds and arbitrary Mellin decay bound the added tail, uniformly on the retained s, z rectangle. Then move this complete w line to $1/2$. Along the move $\Re(s+w) \geq 1+10e$, the numerator is entire, and the denominator is independent of w . The first Euler-region bounds and Mellin decay control the horizontal joins. Both tail errors have arbitrary power saving, including after summing actual rows and prime tuples. This does not extend the buffered reciprocal beyond its authorized s height range. Fubini then gives the full numerator integral inside the two remaining Mellin integrals. Their Gaussian and radial-Mellin profile has uniformly bounded moments, so integrating the numerator estimate introduces no additional contour-height loss. No amplitude indicator is analytically continued.

Rows above $Z^{h+\zeta}$ use the unchanged absolute large-row contour; small and intermediate rows use the comparisons certified above. The normalized probe therefore obeys, for some $0 < \omega < \Delta$ and $\sigma > 0$ independent of the target,

$$|J_\eta(Z)| \ll_\eta Z^{C(B)+\omega}, \quad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma},$$

where

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds.$$

These are the source continuation criterion's hypotheses at B , contradicting $\beta_* > B$. This proves the Hecke conclusion. Restoring finite Euler factors introduces no zeros in this positive half-plane. The norm-pullback factorization $L_F(s, \chi \circ N) = L(s, \chi)L(s, \chi\chi_{-3})$ gives the Dirichlet conclusion exactly as in the source transfer theorem. $\square$

7 Public comparison and limits

The strongest comparable public unconditional claim located by a dated search, completed on 10 October 2026 at 00:16 UTC, is the unique root B_{public} in $(0.8749570194, 0.8749570195)$ of

$$7884B_{\text{public}}^3 - 18819B_{\text{public}}^2 + 14643B_{\text{public}} - 3686 = 0.$$

It is approximately 0.87495701942009894613 . The endpoint appears in written manuscripts of Shunia with GPT-6 Astra [5] and of Guo [6]. A Lean package for it by Nielstron is in an open, unmerged pull request to the QuasiRiemannTracker [7]; its author reports local Comparator and Lean/NanoDa/con-ron checks of seven targets. The best value then catalogued by the tracker, with status framework-verified, is the rational $874957019420098946128604623/10^{27}$ [8]. These reports were read, not rerun. The present $B = 10499/12000$ is smaller than B_{public} by about $4.0353 \cdot 10^{-5}$. This is plain arithmetic:

$$\frac{10499}{12000} = 0.8749166\dots < 0.8749570194 < B_{\text{public}},$$

where 0.8749570194 is the lower endpoint of the interval above. The comparison with the tracker's rational, $10499/12000 < 874957019420098946128604623/10^{27}$, is the Lean theorem `WeightedQRH.theta_lt_public` in `Parameters.lean`, a module in the endpoints' import closure; the proof itself does not use it. The only smaller public value found, Guo's $328/375$ [6], is explicitly conditional on an unproved mean-square hypothesis (Hypothesis 11.1 there). The search is dated and is not an exhaustive novelty or priority certificate. The public records were consulted as benchmarks only; no outside refinement proof code was fetched, used or reproduced here. The new ingredient is the selected-prime saving in a moment of the *physical numerator*, which is absent when one bounds that numerator pointwise and uses moments only to count zero witnesses. The initial unweighted numerator fourth-moment variant did not improve the global comparison; retaining the actual reflected length is essential.

The zero-slot argument uses the stronger certified source theorem, with no restriction on bounded plain lengths; a weaker detector estimate would not suffice there. Exact exponent certificates alone do not verify the analytic deduction; that deduction is checked by the Lean development of Sections 8 and 9. No external human peer review is claimed.

This change invalidates the applicability of the old row-count/pointwise-numerator optimization barrier to the new high bound. It does not remove the unchanged direct estimate's obstruction to GRH. Let g be the extra Gram exponent and e_{row} the extra completed-row energy exponent, distinct from the small contour buffer used above. If they satisfy

$$g \geq b/6, \quad e_{\text{row}} \geq 0, \quad e_{\text{row}} \geq (1 + 3\ell - 2M)/4, \quad e_{\text{row}} \geq M + \ell - 1,$$

then the low comparison requires $B \geq 5/6 + M/12 - \ell/6 + e_{\text{row}}/2$, and

$$\begin{aligned} \frac{M}{12} - \frac{\ell}{6} + \frac{e_{\text{row}}}{2} - \frac{1}{30} &= \frac{1}{5} \left[e_{\text{row}} - \frac{1 + 3\ell - 2M}{4} \right] \\ &+ \frac{1}{60} [e_{\text{row}} - (M + \ell - 1)] + \frac{17e_{\text{row}}}{60} \geq 0. \end{aligned}$$

Thus those particular low estimates still force $B \geq 13/15$. Reaching $1/2$ needs a change on that side as well, such as a proved arithmetic cancellation estimate that survives the reflection phase. This is a limitation of the stated low estimates only, not a lower bound for actual zeros or a theorem excluding other methods.

8 Correspondence with the checked Lean theorems

The new modules are under `formalization/WeightedQRH/` in the proof package. They import the original Lean development [2] at the pin given in Section 9. Some of them, for example the low-side detector and reflection modules, adapt original modules to the changed lengths and record that origin in their headers. The following list maps the steps of the written proof to the theorems composed by the endpoints of `Final.lean` and `Statements.lean`. Apart from the standalone arithmetic barrier `physical_low_barrier`, every theorem named in the list lies on the proof path of one of these endpoints. The names are the actual theorem declarations in the Lean sources, not additional hypotheses standing in for the corresponding proofs. Names from `HighRows/SourceProbe.lean`, `HighRows/WeightedSourceRawCells.lean` and `HighRows/ConditionalFinal.lean` lie in the namespace `OAI.SevenEighths.WeightedHighFinalAssembly`; the others lie in `WeightedQRH` or one of its subnamespaces.

- The low-side comparison at the changed lengths is the actual `sourceProbe_low` theorem in `HighRows/SourceProbe.lean`. The arithmetic geometry and scoped $13/15$ obstruction are in `Parameters.lean`; the obstruction is `physical_low_barrier`, which the proof does not use.
- The polynomial certificates and their real-rectangle implications are in `CertificateBernstein.lean` and `ExponentCertificate.lean`. In particular, `weighted_extended_bound` (with `old_extended_bound` for the source branch) includes the fixed row extension and the $1/20000$ margin. These arithmetic theorems alone imply no zero-free result.
- The masked row count of Section 3 is `masked_amplitude_class_count` in `HighRows/MaskedCount.lean`.
- The combined primitive and redundant conductor bound is `physical_reflection_data_exists` in `NumeratorReflectionData.lean`. The

coefficient mass (8) is `actual_physical_label_mass` in `NumeratorPhysicalPacketMass.lean`; it is stated for a general scale C_u and is applied with this combined conductor.

- On the line $\Re w = 1/2$, each error-label numerator of the complete w integral is identified with its coefficient packet, as in (9), by `errorLabelNumerator_reconstruction` in `NumeratorLabelReconstruction.lean`. The theorems `normalized_prime_tuple_reconstruction` and `integratedDyadValue_reconstruction` in `NumeratorReconstruction.lean` give the complete finite row and error-subset decomposition.
- Proposition 1 is `certified_original_numerator_fourth` in `NumeratorOriginalFourth.lean`. It is proved using the actual certified source moments, rather than assuming a new reflection or fourth-moment estimate.
- The full- w contour transport and uniform remaining Mellin integration are in `FullWContour.lean` through `FullWBound.lean`, including `SourceSZBound.lean`; the transfer from the central cube to the full w line is in `HighRows/CubeFullWTransfer.lean` and `HighRows/CubeFullWNormalized.lean`. The endpoint integration theorem is `FullWContour.finiteFullW_uniform_norm_bound`.
- Final assembly. `Numerator.source_integrated_amplitude_classes` in `NumeratorSourceCollect.lean` collects the actual physical error masks and the original full amplitude classes over all rows of a dyad, with one constant fixed before the row scale and all target, height and contour choices. Combined with the masked count, the scalar, conductor and prime-supply estimates, it proves `Numerator.actual_raw_fullW_input`, of type `RawFullWAnalyticInput`, in `NumeratorActualSource.lean`: the raw full- w bound for every source row dyad. The theorem `fullW_input_of_raw_source` in `HighRows/WeightedSourceRawCells.lean` adds the source geometry, height absorption, remaining Mellin integration and normalization, and yields `FullWAnalyticInput`. This gives `WeightedQRH.fullW_analytic_input` in `Final.lean`.
- The implications `beta_le_theta_of_fullW_input`, `hecke_of_fullW_input`, `dirichlet_of_fullW_input` and `zeta_of_fullW_input` in `HighRows/ConditionalFinal.lean` pass from `FullWAnalyticInput` to the common probe bounds (`common_probe_of_fullW_input`) and through the continuation argument of `Continuation.lean` to the zero-free conclusions. Applied to `WeightedQRH.fullW_analytic_input`, they give the unconditional endpoints `beta_le_theta`, `hecke_nonzero`, `dirichlet_nonzero` and `zeta_nonzero` of `Final.lean`. No hypothesis remains. The docstring of `HighRows/ConditionalFinal.lean` predates `Final.lean` and describes that module in isolation, where `FullWAnalyticInput` is still a hypothesis; the input is discharged in `Final.lean`.
- `Statements.lean` restates the endpoints exactly in the form of the Comparator challenges of Section 9: `OAI.riemannZeta_ne_zero_of_10499_12000_lt_re`, `OAI.DirichletCharacter.LFunction_ne_zero_of_10499_12000_lt_re` and `OAI.SevenEighths.HeckeFamily.LFunction_ne_zero_of_10499_12000_lt_re`.

The endpoint audit `FinalAxioms.lean` prints the statements and the axioms of `actual_raw_fullW_input`, `fullW_analytic_input`, the four endpoints of `Final.lean` and the three statements of `Statements.lean`, and the axioms (not the statement) of `source_integrated_amplitude_classes`. Each of these ten declarations depends only on `propext`, `Classical.choice` and `Quot.sound`. No analytic axiom is introduced anywhere.

9 Formal verification and reproducibility

Proof package. The Lean proof accompanies this manuscript in the proof package `proofs/akashlevy-20261009-weighted-numerator/` prepared for the `QuasiRiemannTracker` [8]. Paths in this section are relative to that directory; Lean file names remain relative to `formalization/WeightedQRH/`. The package contains:

- `formalization/`: a Lake workspace with `lakefile.toml`, `lean-toolchain`, `lake-manifest.json`, the scripts `setup.sh` and `run_comparator.sh`, the 228 Lean modules of the proof in `WeightedQRH/`, the three Comparator challenges with their configurations in `ComparatorChallenges/`, and the three-kernel driver `kernels/judge_three_kernels.py`. The 228 modules are the 227 local modules in the import closure of the endpoint modules `Final` and `Statements`, plus the audit module `FinalAxioms`;
- `manuscript/`: the $\text{\LaTeX}$ sources of this paper;
- `evidence/certificate/`: the exact exponent certificate `weighted_numerator_certificate.py` and its output `weighted_numerator_certificate.json`;
- `evidence/kernels/`: the report, judge log, configuration, export targets, control logs, source audit, tool pins, compile and export logs and the Lean sources `src/Challenge.lean` and `src/Solution.lean` of the three-kernel run, and, in `pristine-workspace/`, the Lake configuration and build log of its challenge workspace;
- `evidence/lake/`: the logs of the Lake build, of its rehash check and of the Comparator runs in that workspace;
- `evidence/isolated-rebuild/` and `evidence/comparator/`: the record of an earlier isolated rebuild and its Comparator runs;
- `evidence/source-scan.txt`, and the license and notice files `LICENSE`, `NOTICE`, `UPSTREAM-NOTICE` and `third-party-notices/`.

Toolchain and source pins. Lean v4.34.1 (commit 5045d0056413266e57c625dcd7c365b10e377c52); Mathlib commit d13f23b723b8a846827a245b89c10fc7d3f11612 [3]; PrimeNumberTheoremAnd commit c39a751132c88b6e8080b74c74023fd95b3d8be0 and rellich-kondrachov commit 70f85d4c1bf99c6e7d61e8be4daa6f3664d08d23, each with the Lean 4.34.1 compatibility patch `lean/patches/*-lean4341.patch` of the original development. All modules are compiled with `autoImplicit` disabled. The original Lean development is `github.com/akashlevy/math` commit c388e88e6519d139ad2ca5571ef0c7fbf0e245b5, branch `qrh-23-24-pruned` [2]. This fork commit is `github.com/openai/math` commit fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb with the declarations that no QRH result uses pruned in place, plus 23/24 modules (`OAI.NumberTheory.DirichletL.QRH*`) that the present proof does not import. The $\text{\LaTeX}$ source of [1] is byte-identical at both commits. One adapted module, `LowDetector/LowOriginalPools.lean`, previously imported an original module that the pruning deleted; it now imports that module’s own imports (`OAI.NumberTheory.DirichletL.Detector.LowSlotScales` and `OAI.NumberTheory.DirichletL.Detector.RayPoolGood`) instead. The three-kernel check additionally uses the official Lean v4.35.0-rc2 release for `darwin_aarch64`. The Challenge and Solution exports of that check, and the Comparator runs in the Lake workspace and on the isolated rebuild, use `lean4export` commit 076e8e57 built with Lean 4.34.1; the preflight

controls of the three-kernel check were compiled and exported with the `lean` and `leanexport` of the v4.35.0-rc2 judge toolchain.

Source scan. The file `evidence/source-scan.txt` records a scan, with comments stripped, of the 231 Lean files in `formalization/` (228 in `WeightedQRH/`, 3 in `ComparatorChallenges/`) for `sorry`, `admit`, `axiom`, `unsafe`, `implemented_by`, `extern`, `skipKernelTC`, `native_decide`, `ofReduceBool`, `trustCompiler`, `run_cmd`, `run_elab`, `run_meta`, `addDecl`, `elab`, `macro_rules`, `macro`, `syntax`, `opaque`, `partial`, `csimp`, `initialize` and `#eval`. No module in `WeightedQRH/` contains any of them. The three challenge modules contain `sorry` by design, since their statements are the targets. The only options set are `maxHeartbeats` and `maxRecDepth`. The scan does not cover the original development or Mathlib; the axiom checks and kernel replays below cover every declaration that the endpoints use.

Lake build from public sources. The script `formalization/setup.sh` clones `github.com/akashlevy/math` at c388e88e6 into `formalization/math/` (a sparse checkout of `lean/OAI/NumberTheory/DirichletL/`, `lean/patches/` and `LICENSE`), lets Lake fetch the dependencies pinned in `lake-manifest.json`, applies the two compatibility patches, downloads Mathlib’s build cache, and runs `lake build WeightedQRH ComparatorChallenges`. In the recorded run, `formalization/math/` was a fresh sparse clone of the fork at that commit. The dependency directory `.lake/packages` was instead an existing checkout of the pinned dependencies with the two patches applied, and the build directory had been seeded with a copy of an earlier build of the original development. The build reported “Build completed successfully (11847 jobs)” after 39 minutes (`evidence/lake/lake-build.log`). Its log lists as built in that run the 2670 modules of the original development in the import closure of the proof, all 228 proof modules and the 3 challenge modules; Mathlib and the other dependencies entered as prebuilt objects of the existing checkout. A subsequent `lake build --no-build --rehash WeightedQRH ComparatorChallenges`, which rehashes the sources and builds nothing, reported “All targets up-to-date (11847 jobs)” (`evidence/lake/lake-no-build-rehash.log`). The output of `FinalAxioms.lean` in the build log shows only `propext`, `Classical.choice` and `Quot.sound` for each of the ten audited declarations (Section 8). Every theorem on the endpoints’ proof paths therefore uses at most these axioms. The recorded run did not exercise the dependency fetch, patch and cache steps of `setup.sh`; its patch step was tested separately on a fresh clone of `rellich-kondrachov` at the pinned revision.

Comparator in the Lake workspace. The Lean FRO Comparator [4] exports a solution theorem with all its dependencies. It checks that the statement and the definitions it uses match the challenge, checks the axioms against the permitted list `propext`, `Quot.sound`, `Classical.choice`, and replays the whole export in the kernel. The challenges in `formalization/ComparatorChallenges/` import only Mathlib. `WeightedQRHZeta` and `WeightedQRHDirichlet` use only Mathlib definitions. `WeightedQRHHecke` is OpenAI’s upstream `HeckeSevenEighths` challenge with only the theorem name and the constant changed. Their solution module is `WeightedQRH.Statements`. The script `formalization/run_comparator.sh` runs `Comparator` (commit d03acab1, default Lean kernel, `NanoDa` disabled) under `lake env` in the workspace above; `Comparator`’s own `lake build` step ran as usual. For each of the three challenges, the log `evidence/lake/comparator-lake-WeightedQRH*.log` shows “Lean default kernel accepts the solution” and “Your solution is okay!”, with exit status 0. Like OpenAI’s upstream 7/8 challenge, the zeta challenge has no hypothesis $s \neq 1$. It holds at $s = 1$ only because Mathlib’s total `riemannZeta` assigns a nonzero value at the pole, which is not an analytic statement there. The analytic statement is `WeightedQRH.zeta_nonzero`.

Three-kernel check of the tracker template. The driver `formalization/kernels/judge_three_kernels.py` was written for this run. It follows the protocol of the tracker’s earlier kernel dossiers (PalomarSubmission `d4e41c1`, driving `lake comparator` from Lean `v4.35.0-rc2`) and re-implements their preflight controls, ill-typed export and export targets; it does not import their script. The challenge is the tracker’s three-target template `public/proofs/palomar-20261009/qrh/src/Challenge.lean` at the catalogue commit of [8], with only the literal `874957019421/1000000000000` replaced by `10499/12000` in all three statements. The resulting `evidence/kernels/src/Challenge.lean` has SHA-256 `4624fc67bbd5792408525558a48481a5cd3acd258c78caf1ec4d29528703c287`. It was compiled with Lean `4.34.1` against an unmodified checkout of `github.com/openai/math` at `fd4aeeb2e` (306 modules of the closure of `Hecke/IdealBridge` built from source) and `Mathlib d13f23b7`. The definition files `Hecke/Family.lean` and `Hecke/IdealBridge.lean` of `openai/math` and `DirichletContinuation.lean` and `RiemannZeta.lean` of `Mathlib` match their pinned git blobs (`evidence/kernels/challenge-source-audit.json`). The module `Solution` (`evidence/kernels/src/Solution.lean`) imports `WeightedQRH.Statements`, compiled in the Lake workspace above, and proves the three template theorems `QRHPalomar.allDirichlet`, `QRHPalomar.zeta` and `QRHPalomar.allHecke` from the endpoints. Both modules were exported with `lean4export` and judged by `lake comparator` of the official Lean `v4.35.0-rc2 darwin_aarch64` release, whose archive digest matches the digest GitHub publishes for that asset (`evidence/kernels/tool-pins.json`). After `Comparator`’s statement, definition and axiom checks, the external kernels `con-ron` (with two jobs) and `NanoDa`, and then Lean’s default kernel, each replayed the export. Before judging, three controls on the same toolchain behaved as expected (`evidence/kernels/control-results.json`): a matching pair was accepted by all three kernels; a mismatched statement was rejected with “Challenge and solution theorem statement do not match”; and an ill-typed proof, a theorem whose value was replaced by its type in the export, was rejected by `con-ron`, `NanoDa` and Lean’s kernel. The judge phase ran on 10 October 2026 from 01:44:59 to 01:53:26 UTC and exited with status 0 (`evidence/kernels/judge.log`). Its log reports “con-ron: accepted 123717 declarations (--verified)”, “con-ron kernel accepts the solution”, “nanoda kernel accepts the solution”, “Lean default kernel accepts the solution” and “Your solution is okay!”. The report `evidence/kernels/result.json` has status `PASS`, verified at 2026-10-10T01:53:26.235107+00:00. The two exports are not published; they were deleted after the run. `Challenge.export` had 382347462 bytes and SHA-256 `f553d092b0690188ea806d9daadef9ec71179fd2d8e22a754dba6e40da9fbacb`; `Solution.export` had 1772535340 bytes and SHA-256 `7ea84555c740c3840fd7f3a3d8cdf1ec987edb60d2825425511c52e6304358a3`. The template’s zeta statement, like the zeta challenge above, has no hypothesis $s \neq 1$.

Earlier isolated rebuild. An earlier route is kept as an additional record in `evidence/isolated-rebuild/` and `evidence/comparator/`. The script `verify_parallel.py` (with `verify.py`, which it imports, and the configuration `lean_environment.json`) rebuilt the 228 proof modules and the 3 challenge modules into a fresh output directory, all with exit status 0, on 10 October 2026 from 00:21:47 to 00:28:17 UTC (`final_rebuild_verification.json`). The live development directory was excluded from the Lean search path, so every local import could resolve only to an object rebuilt in that run. The challenge modules were compiled first, against `Mathlib` and its dependencies only. `Mathlib`, its dependencies and the original development entered as prebuilt objects. The source hashes recorded in the Lake build traces of those prebuilt objects of the original development were compared with the sources at `c388e88e6` and agree; that comparison is not among the evidence files. The same object directory also held stale objects of 253 modules deleted by the pruning; the rebuild admitted an external import only if its module is tracked at `c388e88e6`,

which excludes them. The Lake build above does not depend on these objects. The audit log `WeightedQRH.FinalAxioms.log` of this run shows the same three axioms. With the script `run_comparator.sh` of that directory, Comparator ran against the objects of the isolated rebuild, with its `lake build` step replaced by a shim that only confirms that the prebuilt objects exist. All three challenges printed “Lean default kernel accepts the solution” and “Your solution is okay!”. As a negative control, the zeta challenge with $1/2$ in place of $10499/12000$ was rejected with “Challenge and solution theorem statement do not match” (`evidence/comparator/comparator-NegControl.log`).

Trust caveats. All these checks were run by the author on one macOS arm64 machine. They are not a Palomar registration, an editorial review or a signed tracker receipt, and no independent rerun has been made. The three-kernel check used the official Lean v4.35.0-rc2 `darwin_aarch64` binaries and ran without a sandbox (`lake comparator --inadvisably-no-sandbox`); its log begins with Comparator’s warning “Sandbox disabled, this run is not trustworthy”. The tracker’s earlier kernel dossiers ran on Linux x86_64 inside a bubblewrap sandbox. The driver of this check was written for this run, and its challenge module is named `Challenge` rather than a random alias. In the default-kernel Comparator runs, Comparator’s Landrun sandbox was replaced by an unsandboxed stand-in, and Comparator’s `lean-toolchain` was set to v4.34.1 (pinned v4.34.0) to match the objects; its source is otherwise unchanged. The recorded Lake build did not exercise the dependency steps of `setup.sh`, as stated above. No external human peer review has taken place.

File digests. Every published file of the package is listed with its SHA-256 digest in the adjacent manifest `../akashlevy-20261009-weighted-numerator.sha256.json` (tracker path `proofs/akashlevy-20261009-weighted-numerator.sha256.json`), which lies beside the package directory rather than inside it. Machine-specific path prefixes in the published logs and JSON reports were replaced by neutral prefixes under `/work/qrh-proof/`; `publication-transformations.json` records each replacement and the original digest of every file so changed. The digests quoted above, of `Challenge.lean` and of the two unpublished exports, do not change on publication.

Reproduction. The certificate runs from the package root. The other commands run in `formalization/`, the last two after `setup.sh`. The package stores its files without the executable bit, so the two shell scripts are run with `bash`. The Comparator variables name a Comparator build at the pins above, its `lean4export`, and Landrun or a stand-in. For the three-kernel driver, `<pristine>` is a Lake workspace with the `lakefile.toml` and `lake-manifest.json` of `evidence/kernels/pristine-workspace/`, the `lean-toolchain` of `formalization/`, an unmodified checkout of `github.com/openai/math` at `fd4aeeb2e` in `math/`, and the same two patches applied to its dependencies, built by running `lake build OAI.NumberTheory.DirichletL.Hecke.IdealBridge` in `<pristine>`; `<tracker>` is a checkout of the tracker; `--judge-lean` is the Lean v4.35.0-rc2 release directory, whose `bin/` holds `nanoda_bin` and `con-ron`; and `<fresh>` must not exist.

```
# package root: exact exponent certificate
python3 evidence/certificate/weighted_numerator_certificate.py \
  --output <certificate.json>
# formalization/: Lake build from public sources
cd formalization && bash setup.sh
# formalization/: Comparator, default Lean kernel
export COMPARATOR=<comparator> COMPARATOR_LEAN4EXPORT=<lean4export> \
  COMPARATOR_LANDRUN=<landrun>
for c in Zeta Dirichlet Hecke; do bash run_comparator.sh $c; done
# formalization/: three-kernel check of the tracker template
```

```
python3 kernels/judge_three_kernels.py --solution-workspace . \
  --challenge-workspace <pristine> \
  --template <tracker>/public/proofs/palomar-20261009/qrh/src/Challenge.lean \
  --source-lean <lean-4.34.1> --judge-lean <lean-4.35.0-rc2> \
  --lean4export <lean4export> --out <fresh>
```

The certificate output coincides with `evidence/certificate/weighted_numerator_certificate.json`. The scripts of the earlier isolated route are a record; they expect prebuilt objects at machine-specific locations, set in `lean_environment.json` and in the scripts, which must be adapted before reuse.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\Re s > 7/8$* , 30 September 2026. github.com/openai/math commit `fd4aeeb2e`, preprints/The-Quasi-Riemann-Hypothesis-September-30-2026/. [Pinned manuscript](#).
- [2] OpenAI, Lean development of [1], pruned fork by A. Levy: github.com/akashlevy/math, branch `qrh-23-24-pruned`, commit `c388e88e6519d139ad2ca5571ef0c7fbf0e245b5`, derived from github.com/openai/math commit `fd4aeeb2e`. [Pinned source](#).
- [3] The mathlib Community, *Mathlib*, commit `d13f23b723b8a846827a245b89c10fc7d3f11612`. [Pinned source](#).
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- [5] GPT-6 Astra and J. M. Shunia, manuscript `compensated_half_plane_refinement.pdf`, repository [jshunia/rhtm](#), revision `c3abed0cae8f`. [Pinned manuscript](#).
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- [8] QuasiRiemannTracker (QRH Bounds), maintained by T. Gehringer, catalogue at commit `40844a76`, as of 10 October 2026, 00:16 UTC. [Website](#); [pinned catalogue](#).